



ICE-4101
Artificial Intelligence and Neural Networks

Lecture-1

Concept of Artificial Intelligence

(What is AI; History, Stages, Types of AI)

Course Teacher

Prof. Dr. Md. Ashikur Rahman Khan

Dr. M.A.R. Khan, Dept. of ICE, NSTU

Artificial Intelligence and Neural Networks

	Book List		
	Title	Author(s)	Publisher, Date
1.	Artificial Intelligence A Modern Approach	Stuart J. Russell and Peter Norvig,	Prentice-Hall-2010
2.	Neural Networks, Fuzzy Systems and Evolutionary Algorithms: Synthesis and Applications	S. Rajasekaran, G. A. VijayalakshmiPai	PHI Learning Private Ltd, 2017
3.	Neural Networks: A Comprehensive Foundation	Simon Haykin	Prentice Hall, 1999
4.	An Introduction to Neural Networks	Kevin Gurney	UCL Press, 1997
5.	Neural Networks for Pattern Recognition	Christopher Bishop	Clarendon Press, Oxford, 1995
6.	The Essence of Neural Networks	Robrt Callan	Prentice Hall Europe, 1999
7.	Fundamentals of Neural Networks	Laurene Fausett	Prentice Hall, 1994
8.	Neural Networks and Learning Machines	Simon Haykin	
9.	Neural Networks A Systematic Introduction	Berkeley, California Jerome Feldman	

Dr. M.A.R. Khan, Dept. of ICE, NSTU

Artificial Intelligence and Neural Networks

What will we learn here!

1. What is AI
2. Why AI
3. What are comprises to AI
4. Goals of AI
5. History of AI
6. Types of AI
7. Advantages and Disadvantages of AI
8. Examples of Ai
9. Applications of AI
10. Prerequisites for Artificial Intelligence
11. AI Tools



What is Intelligence?

Intelligence

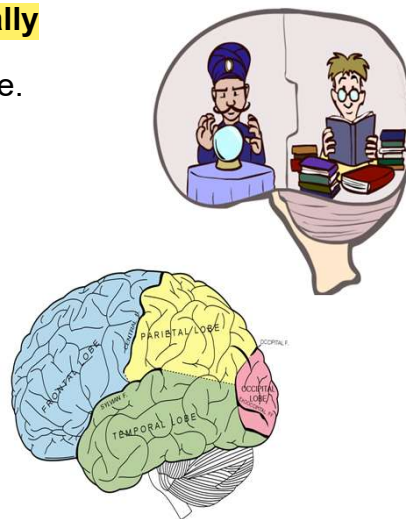
— capacity to learn and solve problems” (Webster dictionary)

— the ability to act rationally

Hmm... Not so easy to define.

What About the Brain?

- Brains (human minds) are very good at making **rational decisions, but not perfect**
- Brains aren't as modular as software, so hard to reverse engineer!
- “**Brains are to intelligence as wings are to flight**”
- Lessons learned from the brain: memory and simulation are key to decision making



Prof. Dr. M.A.R. Khan, Dept. of ICE, NSTU

What is AI?

The science of making machines that:



What is Artificial Intelligence?

- The attempt to make computers more "Intelligent"
- The attempt to better understand human intelligence

Artificial Intelligence is a technique which enables machines to human behavior.

Artificial defines "man-made," and
Intelligence defines "thinking power", hence
AI means "a man-made thinking power."

Computer programs that demonstrate some level of "Human-like" problem solving skills are commonly known as artificial intelligence (AI).

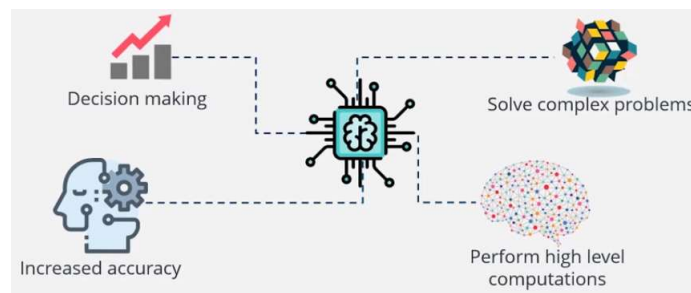
Artificial Intelligence is a branch of computer science that deals with developing intelligent machines which can behave like human, think like human, and has ability to take decisions by their own."



What is Artificial Intelligence?

According to the father of Artificial Intelligence, John McCarthy, AI is *"The science and engineering of making intelligent machines, especially intelligent computer programs"*.

Artificial Intelligence is the development of computer systems that are capable of performing tasks that normally require human intelligence, such as decision making, object detection, solving complex problems, and so on.

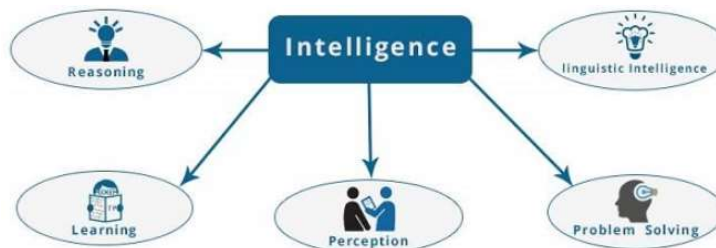


What Comprises to Artificial Intelligence?

What is Intelligence Composed of?

Artificial Intelligence is not just a part of computer science even it's so vast and requires lots of other factors which can contribute to it.

Our brain which is a combination of Reasoning, learning, problem-solving perception, language understanding, etc.



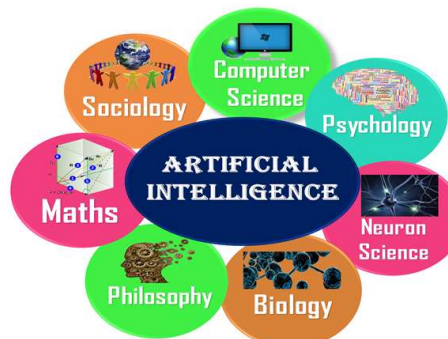
- **Reasoning:** (set of processes that enables us to provide basis for judgement, making decisions, and prediction).
- **Learning:** (activity of gaining knowledge or skill by studying, practicing, being taught, or experiencing something)

What Comprises to Artificial Intelligence?

- **Problem Solving:** also includes **decision making**
- **Perception:** (process of acquiring, interpreting, selecting, and organizing sensory information)
- **Linguistic Intelligence:** (The ability to speak, recognize, and use mechanisms of phonology (speech sounds), syntax (grammar), and semantics (meaning)).

To achieve the above factors for a machine or software Artificial Intelligence requires the following discipline:

- Mathematics
- Biology
- Psychology
- Sociology
- Computer Science
- Neurons Study
- Statistics



Why Artificial Intelligence?

Following are some main reasons to learn about AI:

With the help of AI, you can

- create such software or devices which can solve real-world problems very easily and with accuracy such as health issues, marketing, traffic issues, etc.
- create your personal virtual Assistant, such as Cortana, Google Assistant, Siri, etc.
- build such Robots which can work in an environment where survival of humans can be at risk.
- AI opens a path for other new technologies, new devices, and new Opportunities.

Cortana (Microsoft)

- Voice Commands: Responds to voice commands to perform tasks.
- Reminders and Calendar Management: Sets reminders, schedules meetings, and provides calendar alerts.
- Search and Information: Uses Bing to search the internet and provide answers.
- Productivity Integration: Works with Microsoft Office tools like Outlook, Teams, and OneNote.
- Device Control: Controls Windows settings like Bluetooth, brightness, or launching apps.
- Email Management: Composes, sends, and reads emails aloud.
- Cross-Platform Integration (limited): Some features are available on Android/iOS via Microsoft apps.

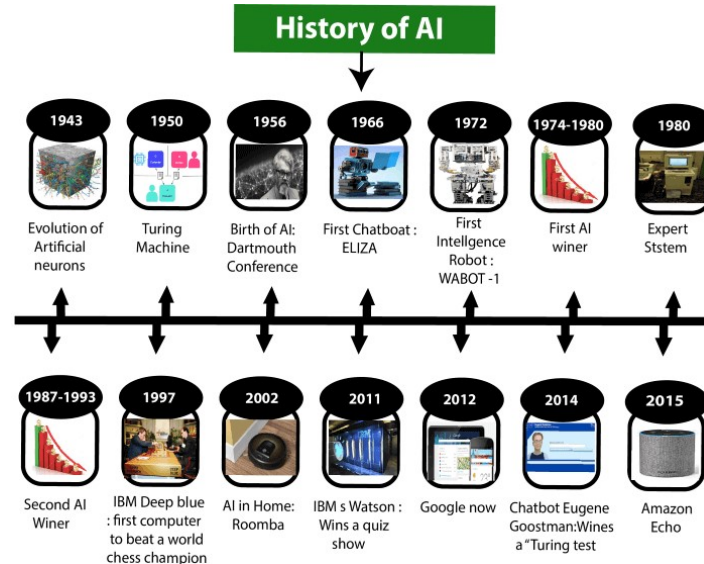
Goals of Artificial Intelligence

Following are the main goals of Artificial Intelligence:

1. Replicate human intelligence
2. Solve Knowledge-intensive tasks
3. An intelligent connection of perception and action
4. Building a machine which can perform tasks that requires human intelligence such as:
 1. Proving a theorem
 2. Playing chess
 3. Plan some surgical operation
 4. Driving a car in traffic
5. Creating some system which can exhibit intelligent behavior, learn new things by itself, demonstrate, explain, and can advise to its user.

History of Artificial Intelligence

AI is not a new word and not a new technology for researchers. This technology is much older than you would imagine. Even there are the myths of Mechanical men in Ancient Greek and Egyptian Myths. Following are some milestones in the history of AI.



History of Artificial Intelligence

Artificial intelligence is assumed a new technology, but in reality, it is not new. The researchers in the field of AI are much older. It is said that the concept of intelligent machines was found in Greek Mythology. Below are some keystones in the development of AI:

Maturation of Artificial Intelligence (1943-1952)

- In the year 1943, **Warren McCulloch** and **Walter Pitts** proposed a model of Artificial neurons.
- In the year 1949: Donald Hebb demonstrated an updating rule also called **Hebbian learning** for modifying the connection strength between neurons.
- In the year 1950, **Alan Turing** published a "Computer Machinery and Intelligence" paper in which he introduced a test, known as a **Turing Test**. This test is used to determine intelligence in machines by checking if the machine is capable of thinking or not.

The birth of Artificial Intelligence (1952-1956)

- In the year 1955: An Allen Newell and Herbert A. Simon created the "first artificial intelligence program" named as "**Logic Theorist**".
- In the year 1956, for the first time, the term AI was coined by the American Computer scientist **John McCarthy** at the Dartmouth Conference. **John McCarthy** is also known as the **Father of AI**.

History of Artificial Intelligence

The golden years-Early enthusiasm (1956-1974)

- In the year 1972, the first intelligent humanoid robot, WABOT1, was created in Japan.

The first AI winter (1974-1980)

- AI winter refers to the time period where computer scientist dealt with a severe shortage of funding from government for AI researches.

A boom of AI (1980-1987)

- In the year 1980, AI came with the evolution of Expert Systems—computer programs designed to solve complex problems.
- In the Year 1980, the first national conference of the American Association of Artificial Intelligence was held at Stanford University.

The second AI winter (1987-1993)

- The duration between the years 1987 to 1993 was the second AI Winter duration.
- Again Investors and government stopped in funding for AI research as due to high cost but not efficient result. The expert system such as XCON was very cost effective.

History of Artificial Intelligence

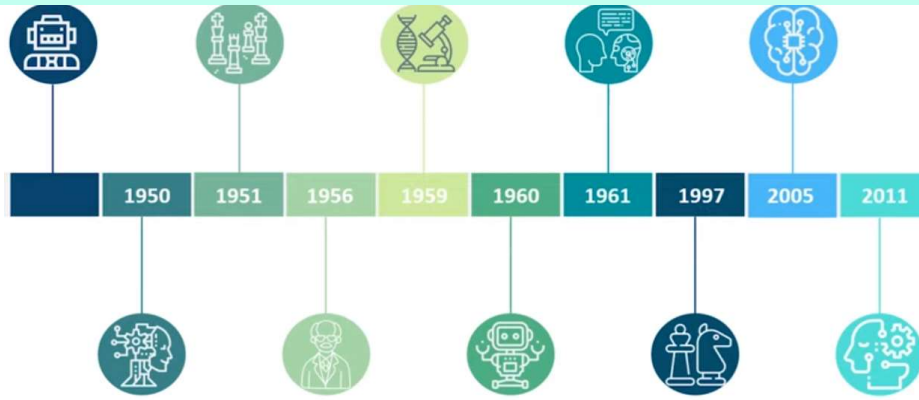
The emergence of intelligent agents (1993-2011)

- In the year 1997, IBM Deep Blue beat world chess champion Gary Kasparov and became the first computer to defeat a world chess champion.
- In the year 2006, AI came into the business world. World's top companies like Facebook, Twitter, and Netflix also started using AI in their applications.

Deep learning, big data and artificial general intelligence (2011-present)

- **Year 2011:** In the year 2011, IBM's Watson won jeopardy, a quiz show, where it had to solve the complex questions as well as riddles. Watson had proved that it could understand natural language and can solve tricky questions quickly.
- **Year 2012:** Google has launched an Android app feature "Google now", which was able to provide information to the user as a prediction.
- **Year 2014:** In the year 2014, Chatbot "Eugene Goostman" won a competition in the infamous "Turing test."
- **Year 2018:** The "Project Debater" from IBM debated on complex topics with two master debaters and also performed extremely well.
- Google has demonstrated an AI program "Duplex" which was a virtual assistant and which had taken hairdresser appointment on call, and lady on other side didn't notice that she was talking with the machine.

Now AI has developed to a remarkable level. The concept of Deep learning, big data, and data science are now trending like a boom. Nowadays companies like Google, Facebook, IBM, and Amazon are working with AI and creating amazing devices. The future of Artificial Intelligence is inspiring and will come with high intelligence.



2011 – IBM Watson

IBM's question answering system, *Watson*, defeated the two greatest Jeopardy! champions, Brad Rutter and Ken Jennings..

Greek Mythology -Talos

Talos was a giant animated bronze warrior programmed to guard the island of Crete created by Hephaestus

1950 – Alan Turing

Alan Turing published a landmark paper in which he speculated about the possibility of creating machines that think.

1951 - Game AI

Christopher Strachey wrote a checkers program and Dietrich Prinz wrote one for chess.

1956 – The birth of AI

John McCarthy first coined the term "Artificial Intelligence" in 1956 at the Dartmouth Conference.

1959 – First AI Laboratory

MIT AI Lab was first set up in 1959. The research on AI began.

1960 – General Motors Robot

First robot was introduced to General Motors assembly line.

1961 – First Chatbot

The first AI chatbot called ELIZA was introduced in 1961.

1997 – IBM Deep Blue

IBM's Deep Blue beats world champion Garry Kasparov in the game of chess.

2005 – DARPA Grand Challenge

Stanford Racing Team's autonomous robotic car, Stanley wins the 2005 DARPA Grand Challenge.

2011 – IBM Watson

IBM's question answering system, *Watson*, defeated the two greatest Jeopardy! champions, Brad Rutter and Ken Jennings..



Turing Test

Turing Test was introduced by British Mathematician Turing in his 1950 paper, "Computing Machinery and Intelligence."

Alan Turing introduced a test to check whether a machine

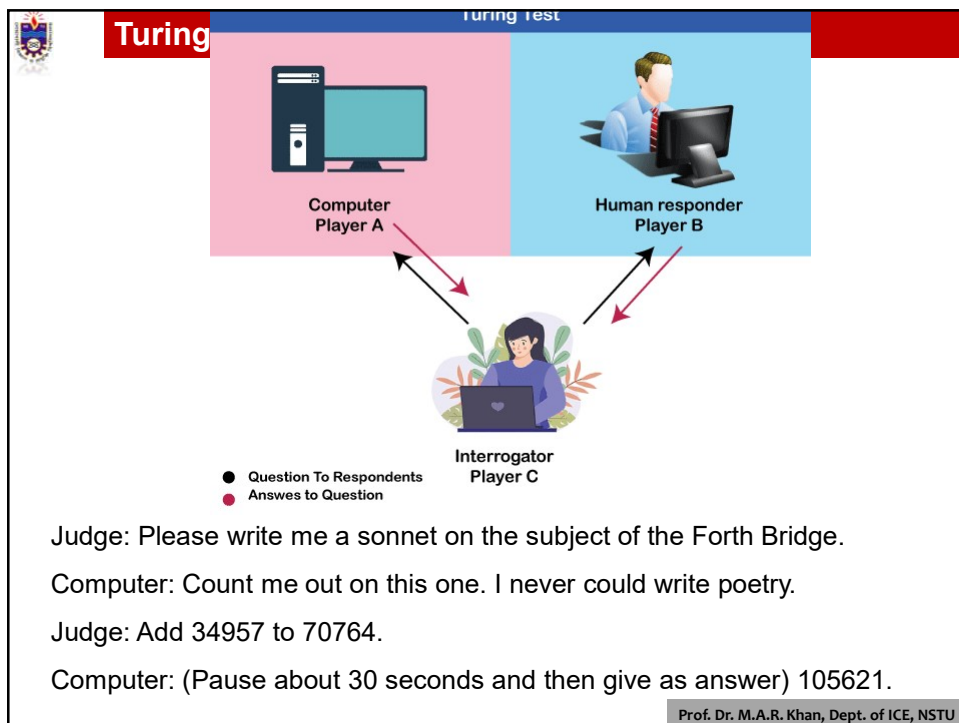
"can think like a human or not?" or

Turing argued that if a machine can pass a certain test ('Turing test') then the **computer is intelligent**. The success of an intelligent behavior of a system can be measured with **Turing Test**. This test is known as the Turing Test.

The Turing test involves a human being (known as the 'judge') asking questions via a computer terminal to two other entities one of which is a human being and the other of which is a computer.

Judge (human) needs to identify this on the basis of questions and their responses. If the judge regularly failed to correctly distinguish the computer from the human, or to identify which is a machine and which is human, then the computer was said to have passed the test successfully.

Prof. Dr. M.A.R. Khan, Dept. of ICE, NSTU



Turing Test (Contd.)

Hi! Are you a computer?

No. My name is Mary.

Are you kidding, I'm Hal and I can't even multiply two-digit numbers!

Judge: Are you a computer? **Computer:** No

Judge: Multiply two large numbers such as (256896489*456725896) **Computer:** Long pause and give the wrong answer.

Prof. Dr. M.A.R. Khan, Dept. of ICE

History of AI

ELIZA:
impersonating a Rogerian psychotherapist

1960s ELIZA Joseph Weizenbaum

<http://www-ai.ijs.si/eliza/eliza.html>

ELIZA - a friend you could never have before

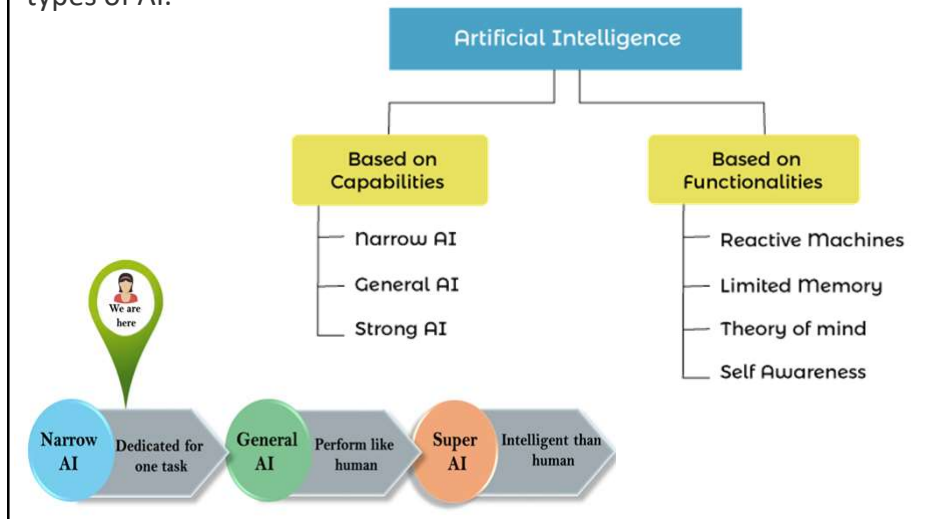
Eliza: Hello. I am ELIZA. How can I help you?

You: Well, I feel sad
Eliza: Do you often feel sad?
You: not very often.
Eliza: Please go on. 😊

Prof. Dr. M.A.R. Khan, Dept. of ICE

Types of Artificial Intelligence:

Artificial Intelligence can be divided in various types, there are mainly two types of main categorization which are based on capabilities and based on functionalities of AI. Following is flow diagram which explain the types of AI.

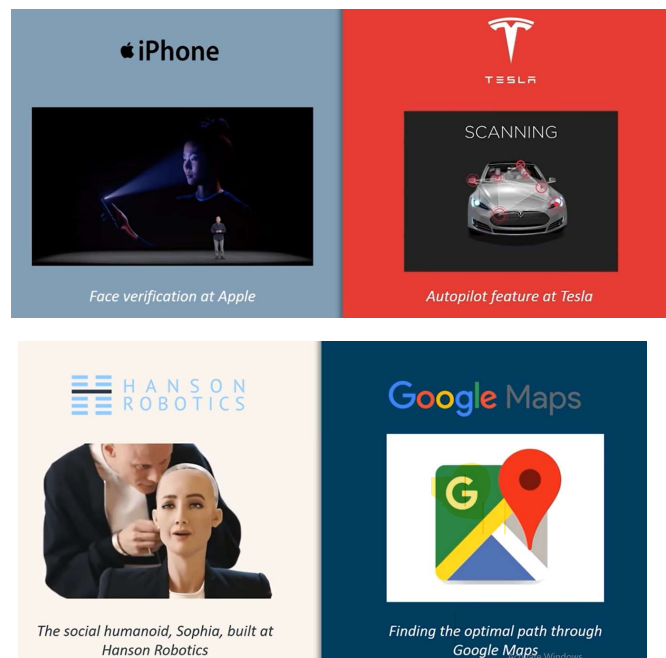


AI type-1: Based on Capabilities

1. Weak AI or Narrow AI:

- Narrow AI is a type of AI which is able to perform a dedicated task with intelligence. The most common and currently available AI is Narrow AI in the world of Artificial Intelligence.
- Apple Siri is a good example of Narrow AI, but it operates with a limited pre-defined range of functions.
- IBM's Watson supercomputer also comes under Narrow AI, as it uses an Expert system approach combined with Machine learning and natural language processing.
- Some Examples of Narrow AI are playing chess, purchasing suggestions on e-commerce site, self-driving cars, speech recognition, and image recognition.

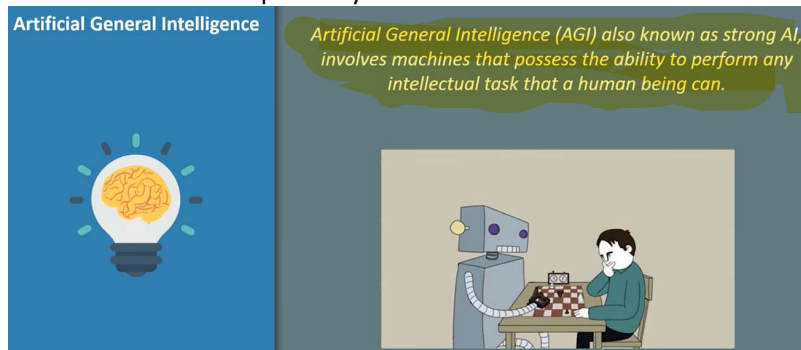
Example of Weak AI or Narrow AI:



AI type-1: Based on Capabilities

2. General AI:

- General AI is a type of intelligence which could perform any intellectual task with efficiency like a human.
- Currently, there is no such system exist which could come under general AI and can perform any task as perfect as a human.
- The worldwide researchers are now focused on developing machines with General AI.
- As systems with general AI are still under research, and it will take lots of efforts and time to develop such systems.



AI type-1: Based on Capabilities

3. Super AI:

- Super AI is a level of Intelligence of Systems at which machines could surpass human intelligence, and can perform any task better than human with cognitive properties. It is an outcome of general AI.
- Some key characteristics of strong AI include capability include the ability to think, to reason, solve the puzzle, make judgments, plan, learn, and communicate by its own.
- Super AI is still a hypothetical concept of Artificial Intelligence. Development of such systems in real is still world changing task.



"Strong AI would take off on its own, and re-design itself at an ever-increasing rate. Humans, who are limited by slow biological evolution, couldn't compete, and would be superseded."

Artificial Intelligence type-2: Based on functionality

1. Reactive Machines

- Purely reactive machines are the most basic types of Artificial Intelligence.
- Such AI systems do not store memories or past experiences for future actions.
- These machines only focus on current scenarios and react on it as per possible best action.
- IBM's Deep Blue system is an example of reactive machines.
- Google's AlphaGo is also an example of reactive machines.



Deep Blue defeated world champion Garry Kasparov (1997) in chess.



That operates solely based on the present data taking into account only the current situation .

Reactive Machine AI includes machines that operate solely based on the present data, taking into account only the current situation. Reactive AI machines cannot form inferences from the data to evaluate their future actions.

Artificial Intelligence type-2: Based on functionality

2. Limited Memory

- Limited memory machines can store past experiences or some data for a short period of time.
- These machines can use stored data for a limited time period only.
- Self-driving cars are one of the best examples of Limited Memory systems. These cars can store recent speed of nearby cars, the distance of other cars, speed limit, and other information to navigate the road.



Self-driving cars use the data collected in the recent past to make immediate decisions.

Limited Memory AI can make informed and improved decisions by studying the past data from its memory. Such an AI has a short-lived or a temporary memory that can be used to store past experiences and hence evaluate future actions.

Artificial Intelligence type-2: Based on functionality

3. Theory of Mind

- Theory of Mind AI should understand the human emotions, people, beliefs, and be able to interact socially like humans.
- This type of AI machines are still not developed, but researchers are making lots of efforts and improvement for developing such AI machines.



Theory of Mind AI is speculated to play a major role in human psychology

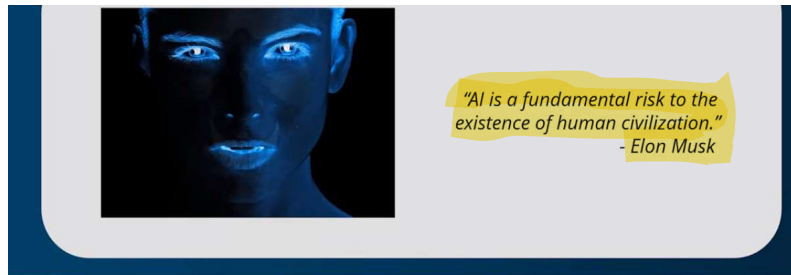
Theory of Mind AI

The Theory of Mind AI will focus mainly on emotional intelligence so that human beliefs and thoughts can be better comprehended.

Artificial Intelligence type-2: Based on functionality

4. Self-Awareness

- Self-awareness AI is the future of Artificial Intelligence. These machines will be super intelligent, and will have their own consciousness, sentiments, and self-awareness.
- These machines will be smarter than human mind.
- Self-Awareness AI does not exist in reality still and it is a hypothetical concept.



Self-Aware AI

The Self-Aware AI includes machines that have their own consciousness and become self-aware. This type of AI does not exist yet.

Advantages of Artificial Intelligence (AI)

Reduction in Human Error

- One of the biggest achievements of AI is that it can reduce human error.

Reduce the Risk (Zero Risk)

- AI can be very helpful in risky situations where humans can't reach or find difficult survival. Such as *defusing a bomb, oil and coal mining and exploring the deepest part of the ocean*, etc. exploring the deepest part of the sea.

24/7 Support

- With AI, 24*7 support can be provided to the customers using chatbots as customer care.

Perform Repetitive Jobs

- Repetitive actions can be boring for human beings, but with AI-enabled machines, they can be performed with full efficiency. Such as *automatic replies to emails, sending birthday and anniversary quotes and verifying documents*, etc.

Advantages of Artificial Intelligence (AI)

Faster decision

Unlike humans, a machine helps to take decisions faster than a human and carry out actions quicker. The best example is an online chess game in the third level. It is impossible to beat a computer machine because it takes the best possible step in a very short time.

New Inventions

For new inventions, AI is helping humans almost in each sector, it can be healthcare, medical, educational, sports, technology, entertainment or research industry etc. Using advanced AI-based technologies, doctors can predict various dangerous diseases like cancer at a very early stage.

Daily Applications

Now, we are all completely dependent on mobile and the internet for our daily routine. We use several applications like Google map, Alexa, Apple's Siri, Window's Cortana, OK Google, taking a selfie, making a phone call, replying to a mail, etc. Further, we can also predict the weather for today and upcoming days with the help of various AI-based methods.

Advantages of Artificial Intelligence (AI)

Digital Assistance

Digital Assistance helps users by gathering previous users queries and providing solutions that users want.

The best example of digital Assistance can be seen on various websites in the form of chatbot support.

A user asks something, and the computer machine provides relevant information like banking, education, travel, and ticket booking sites.

Some chatbots are designed so that it's become hard to determine whether we're chatting with a chatbot or a human being.

Disadvantages of Artificial Intelligence (AI)

High production cost

A computer machine also requires time to time software and hardware updates to meet the latest requirements. Hence, AI also need repairing and maintenance, which need plenty of costs.

Risk of Unemployment

A robot is one of the implementations of Artificial intelligence, and it is replacing jobs and leading to serve unemployment (In some cases). Hence, according to some people, there is always a risk of unemployment because of robots and chatbots instead of humans. For example, in some more technology-oriented countries such as Japan, robots are widely used in manufacturing industries to replace human resources. However, this is not always the truth because as it replaces humans to enhance efficiency, it is also making more jobs opportunities for humans.

Increasing human's laziness

The new inventions of Artificial Intelligence are making humans lazier towards their work, resulting in humans being completely dependent on

Disadvantages of Artificial Intelligence (AI)

machines and robots. If this continues for more upcoming years, then our next generations will become entirely dependent on a machine, resulting in further unemployment and health issues.

Emotionless

We have always learned since childhood that computers or machines don't have emotions. **Humans work like a team, and team management is a key factor for completing a target.** However, there is no doubt that machines are much better when working efficiently, but it is also true that they never replace the human's connection that makes the team.

Lack of creativity

The biggest disadvantage of Artificial Intelligence is its lack of creativity. Artificial Intelligence is a **technology that is completely based on pre-loaded data.**

However, Artificial Intelligence can learn over time with this pre-fed data and past experiences, **but it cannot be creative like humans.**

Disadvantages of Artificial Intelligence (AI)

No Ethics

Ethics and morality are the two most important features of humans, but it isn't easy to incorporate both of these into Artificial Intelligence. AI is rapidly increasing uncontrollably in each sector, so if this continues for the upcoming decades, it may eventually wipe out humanity.

It doesn't have feelings, beliefs, or an internal moral compass.

- No consciousness or intent
- Dependent on training data and rules
- No accountability

No improvement

Artificial Intelligence is a technology completely based on pre-loaded data and experience, so it cannot be improved as human. It can perform the same task repeatedly, but if you want some improvement and changes, you have to change the command for the same. However, it can store unlimited data that humans cannot, but also it cannot be accessed and used like human intelligence.

If inputs are not fed correctly to AI systems, then it may cause harmful results.

Examples of AI-Artificial Intelligence

The following are the examples of AI-Artificial Intelligence:

- 1.Google Maps and Ride-Hailing Applications
- 2.Face Detection and recognition
- 3.Text Editors and Autocorrect
- 4.Chatbots
- 5.E-Payments
- 6.Search and Recommendation algorithms
- 7.Digital Assistant
- 8.Social media
- 9.Healthcare
- 10.Gaming
- 11.Online Ads-Network
- 12.Banking and Finance
- 13.Smart Home devices
- 14.Security and Surveillance
- 15.Smart Keyboard App
- 16.Smart Speaker
- 17.E-Commerce
- 18.Smart Email Apps
- 19.Music and Media Streaming Service
- 20.Space Exploration

How does Siri act?

1. Google Maps and Ride-Hailing Applications

A few years ago, only GPS (satellite-based navigation) was used as a navigation guide. However, artificial intelligence (AI) now provides users with a much better experience in their unique surroundings.



2. Face Detection and Recognition

Utilizing face ID for unlocking our phones and using virtual filters on our faces while taking pictures are two uses of AI that are presently essential for our day-by-day lives.

Face recognition is used in the former, which means that every human face can be recognized. Face recognition is used in the above, which recognizes a particular face.

Examples of AI

Application of AI

Application of AI

Artificial Intelligence has various applications in today's society.

It is becoming essential for today's time because it can solve complex problems with an efficient way in multiple industries, such as Healthcare, entertainment, finance, education, etc. AI is making our daily life more comfortable and fast.

Following are some sectors which have the application of Artificial Intelligence:

Application of AI



Application of AI

Application of AI

Application of AI

1. AI in Astronomy

- Artificial Intelligence can be very useful to solve complex universe problems. AI technology can be helpful for understanding the universe such as how it works, origin, etc.

2. AI in Healthcare

- In the last, five to ten years, AI becoming more advantageous for the healthcare industry and going to have a significant impact on this industry.
- Healthcare Industries are applying AI to make a better and faster diagnosis than humans. AI can help doctors with diagnoses and can inform when patients are worsening so that medical help can reach to the patient before hospitalization.

3. AI in Gaming

- AI can be used for gaming purpose. The AI machines can play strategic games like chess, where the machine needs to think of a large number of possible places.

Application of AI

Application of AI

4. AI in Finance

- AI and finance industries are the best matches for each other. The finance industry is implementing automation, chatbot, adaptive intelligence, algorithm trading, and machine learning into financial processes.

5. AI in Data Security

- The security of data is crucial for every company and cyber-attacks are growing very rapidly in the digital world. AI can be used to make your data more safe and secure. Some examples such as AEG bot, AI2 Platform, are used to determine software bug and cyber-attacks in a better way.

6. AI in Social Media

- Social Media sites such as Facebook, Twitter, and Snapchat contain billions of user profiles, which need to be stored and managed in a very efficient way. AI can organize and manage massive amounts of data. AI can analyze lots of data to identify the latest trends, hashtag, and requirement of different users.

Application of AI

7. AI in Travel & Transport

- AI is becoming highly demanding for travel industries. AI is capable of doing various travel related works such as from making travel arrangement to suggesting the hotels, flights, and best routes to the customers. Travel industries are using AI-powered chatbots which can make human-like interaction with customers for better and fast response.

8. AI in Automotive Industry

- Some Automotive industries are using AI to provide virtual assistant to their user for better performance. Such as Tesla has introduced TeslaBot, an intelligent virtual assistant.
- Various Industries are currently working for developing self-driven cars which can make your journey more safe and secure.

9. AI in Robotics:

- Artificial Intelligence has a remarkable role in Robotics. Usually, general robots are programmed such that they can perform some repetitive task, but with the help of AI, we can create intelligent robots which can perform tasks with their own experiences without pre-programmed.
- Humanoid Robots are best examples for AI in robotics, recently the intelligent Humanoid robot named as Erica and Sophia has been developed which can talk and behave like humans.

Application of AI

Application of AI

10. AI in Entertainment

- We are currently using some AI based applications in our daily life with some entertainment services such as Netflix or Amazon. With the help of ML/AI algorithms, these services show the recommendations for programs or shows.

11. AI in Agriculture

- Agriculture is an area which requires various resources, labor, money, and time for best result. Now a day's agriculture is becoming digital, and AI is emerging in this field. Agriculture is applying AI as agriculture robotics, solid and crop monitoring, predictive analysis. AI in agriculture can be very helpful for farmers.

Application of AI

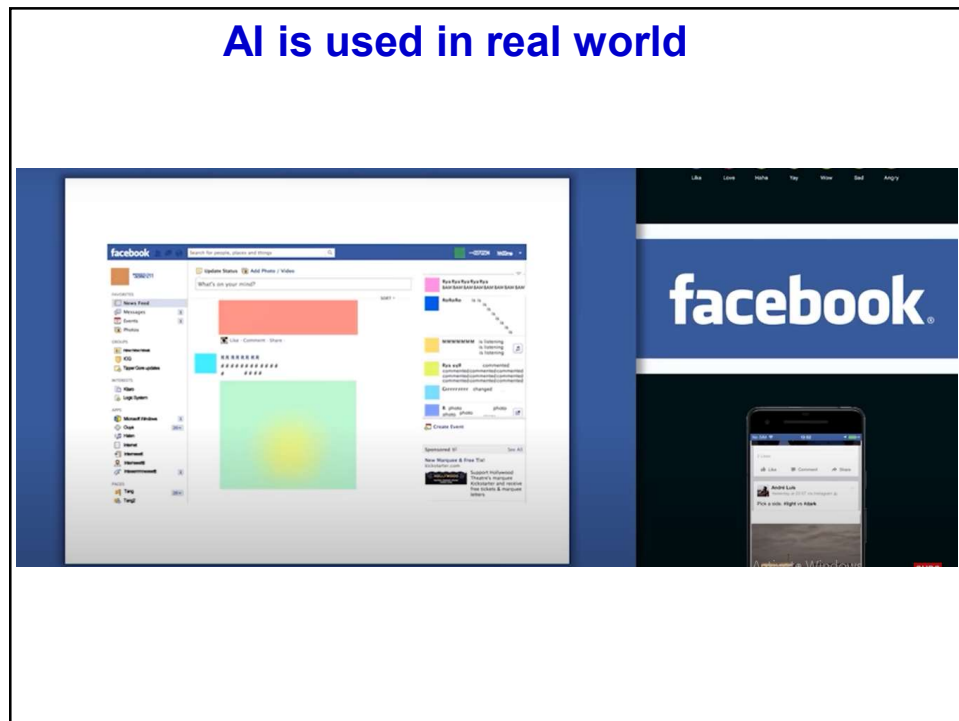
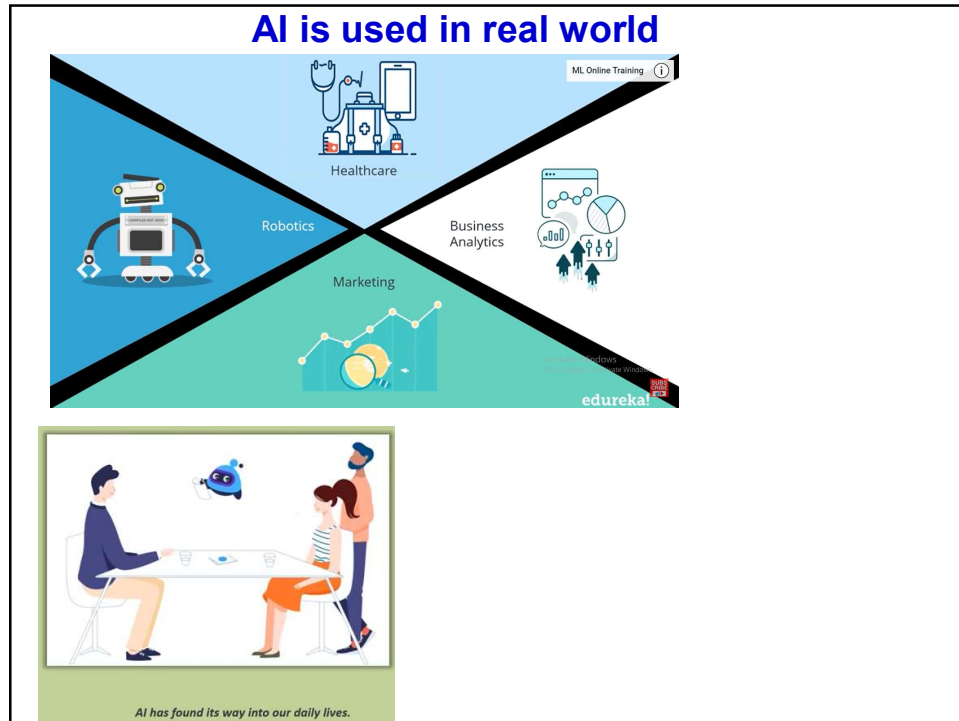
Application of AI

12. AI in E-commerce

- AI is providing a competitive edge to the e-commerce industry, and it is becoming more demanding in the e-commerce business. AI is helping shoppers to discover associated products with recommended size, color, or even brand.

13. AI in education:

- AI can automate grading so that the tutor can have more time to teach. AI chatbot can communicate with students as a teaching assistant.
- AI in the future can be work as a personal virtual tutor for students, which will be accessible easily at any time and any place.



AI is used in real world

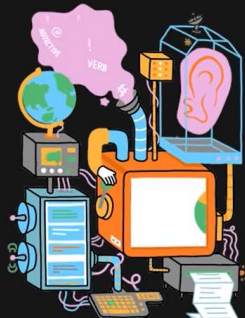


AI is used in real world



J.P.Morgan

JPMorgan Chase's Contract Intelligence (COiN) platform uses AI, machine learning and image recognition software to analyze legal documents.





Manually reviewing 12,000 agreements takes over 36,000 hours

Marketing

Health care

AI is used in real world

Millions of Healthcare organizations use IBM AI (Watson) technology for medical diagnosis.



IBM is one of the pioneers that has developed AI software perfectly for medicine.

More than 230 healthcare organizations worldwide use IBM Watson technology.



In 2016 IBM Watson AI technology was able to cross-reference 20 million oncology records correctly diagnose a rare leukaemia condition in a patient.

Prerequisites for Artificial Intelligence

Prerequisite

Before learning about Artificial Intelligence, you must have the fundamental knowledge of following so that you can understand the concepts easily and that will help you to get started with AI technology.

- Strong knowledge of Mathematics mainly, Calculus, Linear Algebra, Statistics and probability.
- Knowledge of essential Mathematics such as derivatives, probability theory, etc.
- Any computer language such as C, C++, Java, Python, etc.(knowledge of Python will be an advantage). A good experience in programming languages like Java, Python, R.
- A strong understanding of algorithms.
- Good background in data analytics skills.

AI Tools

The list of the most commonly used tools and frameworks for artificial intelligence is provided here.

1. **Scikit Learn**
2. **TensorFlow**
3. **Theano**
4. **Keras**
5. **Caffe**
6. **MxNet**
7. **Google ML Kit**

The more details can be found link below

AI Tools



THANKS